

Wave 2.2 Harmonic Temporal Hybrid Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved wave 2.2 harmonic-temporal hybrid campaign. The campaign tested explicit sparse RCIM harmonic features inside temporal convolution, GRU , and LSTM sequence regressors.

The campaign completed all 9 planned runs with zero launcher failures.

Campaign Artifacts

Artifact	Path
Campaign output	output/training_campaigns/2026-05-25-15-44-37_wave2b_harmonic_temporal_hybrid_campaign_2026_05_25
Campaign leaderboard	output/training_campaigns/2026-05-25-15-44-37_wave2b_harmonic_temporal_hybrid_campaign_2026_05_25/campaign_leaderboard.yaml
Best-run pointer	output/training_campaigns/2026-05-25-15-44-37_wave2b_harmonic_temporal_hybrid_campaign_2026_05_25/campaign_best_run.yaml
Execution report	output/training_campaigns/2026-05-25-15-44-37_wave2b_harmonic_temporal_hybrid_campaign_2026_05_25/campaign_execution_report.md
Planning report	doc/reports/campaign_plans/wave_2/2026-05-25-13-34-12_wave2b_harmonic_temporal_hybrid_campaign_plan_report.md
Model technical document	doc/technical/2026-05/2026-05-25/2026-05-25-03-17-26_wave2b_harmonic_temporal_hybrids.md
Closeout workflow document	doc/technical/2026-05/2026-05-26/2026-05-26-14-01-40_campaign_closeout_and_manual_track2_gate.md

Execution Summary

Field	Value
Campaign name	wave2b_harmonic_temporal_hybrid_campaign_2026_05_25
Started at	2026-05-25T15:44:37
Finished at	2026-05-25T20:05:38
Completed runs	9
Failed runs	0
Tested model types	periodic_temporal_convolution , periodic_gru_sequence , periodic_lstm_sequence
Tested direction scopes	global , Fw , Bw
Harmonic feature list	[0, 1, 3, 39, 40, 78, 81, 156, 162, 240]

Campaign Winner

Field	Value
Run name	te_periodic_gru_sequence_remote_Bw
Run instance	2026-05-25-17-38-18__te_periodic_gru_sequence_remote_bw
Model family	periodic_gru_sequence_bw
Model type	periodic_gru_sequence
Direction scope	backward only
Trainable parameters	157,953
Validation MAE	0.002523
Validation RMSE	0.002965
Test MAE	0.002344
Test RMSE	0.002747

The winning checkpoint is stored at `output/training_runs/periodic_gru_sequence_bw/2026-05-25-17-38-18__te_periodic_gru_sequence_remote_bw/checkpoints/periodic_gru_sequence-epoch=252-val_mae=0.00252321.ckpt` .

Leaderboard

Rank	Run	Family	Scope	Test MAE	Test RMSE	Val MAE	Params
1	te_periodic_gru_sequence_remote_Bw	periodic_gru_sequence_bw	Bw	0.002344	0.002747	0.002523	157,953
2	te_periodic_lstm_sequence_remote_Bw	periodic_lstm_sequence_bw	Bw	0.002556	0.002953	0.002432	210,561
3	te_periodic_gru_sequence_remote_global	periodic_gru_sequence	global	0.002681	0.002971	0.002507	157,953
4	te_periodic_lstm_sequence_remote_global	periodic_lstm_sequence	global	0.002682	0.002969	0.002526	210,561
5	te_periodic_gru_sequence_remote_Fw	periodic_gru_sequence_fw	Fw	0.003193	0.003583	0.003227	157,953
6	te_periodic_lstm_sequence_remote_Fw	periodic_lstm_sequence_fw	Fw	0.003274	0.003651	0.003254	210,561
7	te_periodic_temporal_convolution_sequence_remote_Fw	periodic_temporal_convoluti on_fw	Fw	0.003337	0.003830	0.003321	158,529
8	te_periodic_temporal_convolution_sequence_remote_global	periodic_temporal_convoluti on	global	0.003508	0.003929	0.003634	158,529
9	te_periodic_temporal_convolution_sequence_remote_Bw	periodic_temporal_convoluti on_bw	Bw	0.003614	0.004163	0.003890	158,529

Technical Interpretation

The strongest Wave 2.2 result is the backward-only `periodic_gru_sequence` surface, with test MAE 0.002344. The two recurrent hybrid families dominate the periodic temporal-convolution family in this campaign, and the global `periodic_gru_sequence` and `periodic_lstm_sequence` results are almost tied on scalar test error.

The campaign is a clear improvement over the first Wave 2.1 temporal-sequence entry campaign on the same scalar training-registry metric surface. The best first-wave temporal entry campaign result was

`te_gru_sequence_remote_Fw` at test MAE `0.003333`, while Wave 2.2 reaches `0.002344` on the backward-only periodic GRU surface and `0.002681` on the global periodic GRU surface.

This closeout does not promote the Wave 2.2 winner as the official deployed or accepted TE Curve Verification Pipeline baseline. Campaign metrics are training-registry metrics; the direction-aware offline curve matrix and visual overlays remain a separate approval step. The interrupted TE Curve Verification Pipeline attempt produced no valid result artifact and was removed from the verification surface.

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
<code>periodic_temporal_convolution</code>	<code>te_periodic_temporal_convolution_sequence_remote_global</code>	0.003508	Updated by this campaign
<code>periodic_temporal_convolution_fw</code>	<code>te_periodic_temporal_convolution_sequence_remote_Fw</code>	0.003337	Updated by this campaign
<code>periodic_temporal_convolution_bw</code>	<code>te_periodic_temporal_convolution_sequence_remote_Bw</code>	0.003614	Updated by this campaign
<code>periodic_gru_sequence</code>	<code>te_periodic_gru_sequence_remote_global</code>	0.002681	Updated by this campaign
<code>periodic_gru_sequence_fw</code>	<code>te_periodic_gru_sequence_remote_Fw</code>	0.003193	Updated by this campaign
<code>periodic_gru_sequence_bw</code>	<code>te_periodic_gru_sequence_remote_Bw</code>	0.002344	Updated by this campaign
<code>periodic_lstm_sequence</code>	<code>te_periodic_lstm_sequence_remote_global</code>	0.002682	Updated by this campaign
<code>periodic_lstm_sequence_fw</code>	<code>te_periodic_lstm_sequence_remote_Fw</code>	0.003274	Updated by this campaign
<code>periodic_lstm_sequence_bw</code>	<code>te_periodic_lstm_sequence_remote_Bw</code>	0.002556	Updated by this campaign

The program-level training registry now points to `te_periodic_gru_sequence_remote_Bw` as the scalar best training result. That registry state is not the same thing as official TE Curve Verification Pipeline acceptance.

TE Curve Verification Pipeline Boundary

TE Curve Verification Pipeline was not completed as part of this closeout. Under the updated campaign governance rule, a future TE Curve Verification Pipeline refresh must be prepared as a separate operator-launched PowerShell workflow with local and `-Remote` execution modes. Codex should provide the launcher and exact command, then wait for the operator to run it and report completion.

Closeout Decision

The campaign is complete and successful from an execution standpoint: all 9 runs completed, the leaderboard and best-run artifacts exist, and the family registries were refreshed.

From a modeling standpoint, the periodic recurrent sequence families are worth keeping as verified training candidates. The backward-only periodic GRU is the scalar campaign winner, while the global periodic GRU and periodic LSTM are the strongest bidirectional candidates to carry into a future optional TE Curve Verification Pipeline review.

Recommended Follow-Up

1. Review and commit the normal closeout package if the report and PDF are accepted.
2. Prepare the separate operator-run TE Curve Verification Pipeline launcher only after explicit approval.
3. Use the future TE Curve Verification Pipeline matrix and visual reports to decide whether Wave 2.2 changes any official baseline, rather than using this campaign leaderboard alone.